

OSSAMA TAHA

BI & DATA ANALYTICS ENGINEER

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SUMMARY

Business Intelligence & Data Analytics specialist with a B.Com in Accounting and 2+ years of experience transforming raw data into actionable insights through automated reporting systems, executive dashboards, and data-driven decision frameworks. Expert in designing BI solutions that bridge technical implementation with business impact, reducing reporting cycles by 10x while enabling sales and operations teams to make faster, informed decisions. Combines strong business acumen with technical expertise in SQL, Python, Power BI, and modern data stack technologies.

EXPERIENCE

Nile for Cables and Electronics, BI & Sales Data Analyst

- Reduced reporting cycle from days to hours by designing and deploying a semi-realtime automated reporting system, directly improving sales team responsiveness and strategic decision-making
- Led quarterly sales performance evaluations using consolidated KPIs, providing actionable insights that drove revenue-focused business decisions
- Built executive dashboards supporting sales forecasting and real-time KPI monitoring, enabling stakeholders to track performance without manual data pulls
- Designed end-to-end ETL pipelines using PostgreSQL and Dolt (version-controlled SQL database), ensuring data integrity and audit trail for critical financial metrics
- Developed a local AI chatbot using open-source LLMs for natural language querying of business data, empowering non-technical stakeholders to self-serve analytics

Giza, Egypt
Nov 2024 – present
1 year 7 months

Samsung Innovation Campus, Data Engineer

- Developed scalable data processing workflows using Apache Spark and Kafka, supporting business intelligence and analytical use cases
- Performed exploratory data analysis and feature engineering to enable data-driven business insights
- Deployed production data workflows on AWS cloud infrastructure (EC2, S3), ensuring reliable and cost-efficient operations

Giza, Egypt
July 2024 – Oct 2024
4 months

PROJECTS

dbt Data Warehouse — Star Schema & Analytics Layer

- **Problem:** Raw operational data lacked structure for analytics, forcing manual data wrangling for every report
- **Process & Solution:** Designed star schema with 15+ dbt models across staging/marts layers, SCD Type 2 tracking, incremental processing, and 99.9% data quality via automated testing
- **Business Impact:** Reduced query complexity by 70%, eliminated manual data prep, enabled self-service analytics

Atlas Analytics Platform

- **Problem:** Siloed data across departments required manual updates with no unified visibility
- **Process & Solution:** Built full-stack platform with Python backend, automated ETL pipelines, role-based dashboards, deployed on AWS with CI/CD
- **Business Impact:** Enabled data-driven decisions across 3 departments, reduced manual reporting by 80%

Airflow ETL Pipeline — Multi-Source Data Orchestration

- **Problem:** Manual API data extraction was error-prone with no scheduling or failure recovery
- **Process & Solution:** Designed 4 production DAGs with task dependencies, dynamic mapping, retry logic, extracting from 3 API sources into PostgreSQL with real-time monitoring
- **Business Impact:** Fully automated daily/weekly ingestion with zero manual intervention

Agentic AI Workflow — Multi-Agent DataOps Automation

- **Problem:** Data operations required constant manual monitoring and debugging across multiple systems
- **Process & Solution:** Built multi-agent system with orchestrator pattern, 5 specialized agents, tool integration, and persistent memory using LangChain/LangGraph
- **Business Impact:** Cut pipeline failure resolution time by 70%, automated 90% of routine monitoring

AWS Cloud Data Pipeline — Serverless ETL with IaC

- **Problem:** On-premise infrastructure lacked scalability and cost efficiency for growing data
- **Process & Solution:** S3 data lake + Glue ETL + Athena queries + Step Functions orchestration, fully deployable via Terraform IaC
- **Business Impact:** 90% cost reduction vs traditional infra, sub-second query performance, zero server management

Kafka Streaming Pipeline — Real-Time Data Processing

- **Problem:** Batch processing couldn't meet real-time analytics needs for stock monitoring and telemetry
- **Process & Solution:** Event-driven architecture with Kafka producers, consumer groups, windowed aggregation, dead letter queues, live dashboard, and PostgreSQL sink
- **Business Impact:** Sub-second data latency, automated threshold alerting, real-time operational visibility

SKILLS

BI & Data Visualization: Power BI, DAX, Tableau, Excel, Streamlit, Plotly, Dash

Data Analysis & Interpretation: Statistical Analysis, Data Mining, Business Insights, KPI Development, Trend Analysis, Predictive Modeling

Data Pipeline & Architecture: ETL/ELT Pipelines, Apache Airflow, dbt, Apache Spark, Kafka, Data Modeling (Star Schema), PostgreSQL, MySQL

Cloud & Infrastructure: AWS (S3, Glue, Athena, Lambda, Step Functions, EC2), Terraform, Docker, Docker Compose, Linux Server Administration, Nginx, GitHub Actions (CI/CD)

Programming & Scripting: Python, SQL, Bash, JavaScript, TypeScript, HTML/CSS

AI & Machine Learning: LangChain, LangGraph, Hugging Face Transformers, Scikit-learn, TensorFlow, NLP, Local LLM Deployment

Data Quality & Monitoring: Great Expectations, Data Validation, Schema Testing, Anomaly Detection, Pipeline Monitoring

Automation & Tools: Git, n8n, OpenClaw, KiloCode, Hermes, Dolt, Jupyter, VS Code

EDUCATION

Al-Azhar University , Accounting & Business Management	Sept 2019	–	June 2023
Bachelor of Commerce			
• Self-taught in programming, data engineering, and cloud technologies through professional certifications and hands-on projects • Coursework in financial accounting, cost accounting, managerial accounting, business statistics, and financial statement analysis			

CERTIFICATIONS

Google Data Analytics Professional Certificate

- Data cleaning, analysis, visualization, and storytelling using spreadsheets, SQL, Tableau, and R

IBM Data Engineer Professional Certificate

- ETL pipelines, data warehousing, big data technologies (Spark, Hadoop, Kafka), and cloud deployment

IBM Data Science Professional Certificate

- Machine learning, data analysis, Python, SQL, and applied data science capstone projects

Samsung Innovation Campus Big Data Engineering Certificate

- **Top Achiever (#1)** in the Big Data Engineering track
- Intensive training in big data infrastructure, distributed computing, and data engineering fundamentals

AWS Certified Data Engineer – Associate

- AWS data engineering stack – S3, Glue, Athena, Lambda, Step Functions, and Terraform IaC (In Progress)

Master of Business Administration (iMBA)

- University of Illinois Gies College of Business
- Program includes specializations in Business Analytics, Financial Management, Strategic Leadership, and Managerial Economics

LANGUAGES

Languages: Arabic: Native | English: Professional | Russian: Basic